

- ① For a transistor, $\beta = 45$ and voltage drop across $1k\Omega$ resistor connected in the collector circuit is 1V. Find the base current for common emitter configuration.

$$\beta = 45, R_c = 1k\Omega, V_c = 1\text{V}$$

$$I_c = \frac{V_c}{R_c} = \frac{1\text{V}}{1k\Omega} = 1\text{mA}$$

$$\beta = \frac{I_c}{I_B}$$

$$I_B = \frac{I_c}{\beta} = \frac{1\text{mA}}{45} = \underline{\underline{0.022\text{mA}}}$$

- ② For a common base configuration, the current gain is 0.9. If $I_E = 1\text{mA}$, find the value of I_B .

$$\alpha = \frac{I_c}{I_E} \Rightarrow I_c = (\alpha)(I_E)$$

$$I_c = 0.9 \times 1\text{mA} = 0.9\text{mA}$$

$$I_E = I_B + I_c$$

$$I_B = I_E - I_c = 1\text{mA} - 0.9\text{mA}$$

$$I_B = \underline{\underline{0.1\text{mA}}}$$

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(2)
③ A transistor in CE configuration is working as switch with following values

$$V_{cc} = 5V, V_{CE(sat)} = 0.2V, R_c = 1.2k\Omega$$

$$\beta = 120$$

Determine minimum ^{base} current needed to close the switch.

$$V_{ce} = V_{CE} - I_c R_c$$

$$I_c = \frac{V_{cc} - V_{CE}}{R_c} = \frac{(5 - 0.2)V}{1.2} = \frac{4.8V}{1.2k\Omega} = 4mA$$

$$I_B = \frac{I_c}{\beta} = \frac{4mA}{120} = 0.0333mA$$

$$I_B = 33.3\mu A$$

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- ④ The current gain for CB configuration is 0.95. The voltage drop across $2\text{ k}\Omega$ resistance connected to the collector is 2V. What is base current.

$$\alpha = 0.95, R_C = 2\text{ k}\Omega, V_C = 2\text{ V}$$

$$I_C = \frac{V_C}{R_C} = 1\text{ mA}$$

$$I_E = I_C + I_B$$

$$\alpha = \frac{I_C}{I_E} \Rightarrow I_E = \frac{1\text{ mA}}{0.95} = 1.050\text{ mA}$$

$$\therefore I_B = 1.050 - 1 = \underline{\underline{0.05\text{ mA}}}$$

- ⑤ A transistor is connected in CE in which collector supply is 8V and voltage across the resistance ^{of 800Ω} connected to collector is 0.5V. The value of $\alpha = 0.96$. Determine

(a) V_{CE} (b) I_B

$$R_C = 800\Omega, V_C = 8\text{ V}, I_C = \frac{0.5}{800} = \frac{5}{8}\text{ mA} = 0.625\text{ mA}$$

$$V_{CE} = V_{CC} - 0.5 = 8 - 0.5 = 7.5\text{ V}$$

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.96}{1-0.96} = 24$$

$$\therefore \beta = \frac{I_C}{I_B}, I_B = \frac{I_C}{\beta} = \frac{0.625}{24} = \underline{\underline{0.026\text{ mA}}}$$

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